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JD-30, an active fraction extracted from Danggui–Shaoyao–San, decreases β-amyloid content and deposition, improves LTP reduction and prevents spatial cognition impairment in SAMP8 mice

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ABSTRACT

JD-30 is an active fraction extracted from Danggui–Shaoyao–San (DSS), a traditional Chinese medicinal prescription. We previously showed that JD-30 could alleviate cognitive dysfunction of the mice induced by intracerebroventricular injection of β-amyloid (Aβ). However, data remain scarce on the effect of JD-30 on an Alzheimer's disease (AD) model and the underlying mechanisms are unknown. Further detailed studies on the effects of JD-30 on spatial cognition of senescence-accelerated mouse prone 8 (SAMP8), a suitable rodent model for cognitive impairment of aged subjects were investigated to elucidate the possible mechanisms. Long-term treatment with JD-30 significantly decreased the prolonged latency of SAMP8 in the Morris water-maze test. It also ameliorated the reduction of long-term potentiation (LTP) and reduced the damage of neurons in the hippocampus of SAMP8. Finally, JD-30 decreased the content and deposition of Aβ in the brain of SAMP8. The results show that JD-30 improves deterioration of spatial learning and memory in the SAMP8 mouse model, and by decreasing the content and deposition of Aβ, neuronal activity and synaptic plasticity improve, suggesting one of the mechanisms involved.

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1. Introduction

Alzheimer's disease (AD) is one of the most important global healthcare problems (Cummings, 2004). The number of patients with AD will probably increase from ~5.3 million in 2009 to 11–16 million by 2050 (Alzheimer's Association, 2009). Exploring the pharmacotherapy is a constant and significant topic in medical field. While the debate continues, β-amyloid (Aβ) is believed as the pivotal molecule in the pathogenesis of AD, and accumulation of Aβ in the brain may be critical in the onset of AD (Hardy and Selkoe, 2002).

Danggui–Shaoyao–San (DSS), also called Toki-shakuyaku-san or TJ-23, is a traditional Chinese medicinal prescription, widely used in China, Japan, and Korea (Fujii, 2002; Hagino, 1994; Higaki et al., 2002; Inanaga, 2007; Kotani et al., 1997). DSS was recorded in the 'Synopsis of Prescriptions of the Golden Chamber', which was compiled by Zhong-Jing Zhang during the Han dynasty. The prescription was used to relieve menorrhalgia and other abdominal pains in women. In the 1980's, the therapeutic effect of DSS on AD was first reported by researchers in Japan (Mizushima, 1989), but the underlying mechanisms and substance basis of DSS on AD remain unknown.

Through a serial of preliminary screening and pharmacological testing, JD-30 looked to be a promising active fraction from several extracted fractions of DSS (Hu et al., 2007, 2010). Previous studies had shown that JD-30 could improve spatial cognition deficit in mice induced by intracerebroventricular injection of Aβ fragment 25–35 ($A\beta_{25-35}$), and reverse long-term potentiation (LTP) reduction in hippocampal slices induced by $A\beta_{25-35}$. Blocking and disrupting aggregation of $A\beta_{25-35}$ could be the underlying mechanisms involved. However, research has so far failed to detect Aβ deposition in the brain of the mouse. Meanwhile, data remain scarce on the effect of JD-30 on AD model mouse, making more detailed studies are necessary.

Senescence-accelerated mouse prone 8 (SAMP8) is an autogenic senile strain of mouse characterized by age-related deteriorations, such as loss of memory and retention at an early age; it also produces increased amounts of amyloid precursor protein (APP) and Aβ similar to those observed in AD patients (Butterfield and Poon, 2005; Del

Abbreviations: Aβ, β-amyloid; AD, Alzheimer's disease; APP, amyloid protein precursor; BACE1, beta-site amyloid cleave enzyme-1; CatB, cathepsin B; DG, dentate gyrus; DSS, Danggui–Shaoyao–San; GAPDH, glyceraldehyde phosphate dehydrogenase; HFS, high frequency stimulus; IDE, insulin degrading enzyme; IOD, integrated optical density; LTP, long-term potentiation; NEP, neprilysin; PCR, polymerase chain reaction; PS, population spike; SAMP8, Senescence-accelerated mouse prone-8.

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Valle et al., 2010; Yagi et al., 1988). The features make SAMP8 as a good model to investigate the fundamental mechanisms of age-related learning and memory deficits, and to evaluate the action of drugs.

We have investigated the effect of JD-30 on spatial cognitive performance of SAMP8, and tried to elucidate the underlying mechanisms. First, the spatial learning and memory ability of SAMP8 was investigated after long-term administration (8 weeks) of JD-30. As the neurobiological and anatomical basis of learning and memory, LTP induction and Nissl-positive cell densities in hippocampus were examined. The content and deposition of Aβ were followed by histopathology methods. Lastly, the expression of 5 genes related to the production and clearance of Aβ was analyzed by real-time polymerase chain reaction (PCR).

2. Materials and methods

2.1. Preparation of DSS and JD-30

DSS is composed of the following 6 raw materials: *Angelica sinensis* (Oliv.) Diels (Umbelliferae), *Paeonia lactiflora* Pall. (Ranunculaceae), *Ligusticum chuanxiong* Hort. (Umbelliferae), *Poria cocos* (Schw.) Wolf (Polyporaceae), *Atractylodes macrocephala* Koidz. (Compositae) and *Alisma orientalis* (Sam.) Juzep. (Alismaceae). These materials purchased from Tongrentang Pharmaceutical Company (Beijing, China) were authenticated by Dr. Y.M. Zhao and Dr. Q.Y. Ma, both are botanists in the Department of Phytochemistry in our institute. The voucher specimens were deposited in the Department of Phytochemistry, Beijing Institute of Pharmacology and Toxicology.

The 6 raw materials were mixed in the dry weight ratio of 3:16:8:4:4:8, and the total dry weight was 516 g. The mixture were left in 95% ethanol (1:5 w/v) overnight at room temperature, and boiled twice for 2 h each time. After filtration and centrifugation, the extract was concentrated and referred to as DSS-A (53.5 g, 10.36%, w/w). The residue was boiled with distill water twice for 1 h each time, and filtered to obtain the filtrate. The filtrate was concentrated and lyophilized to obtain the preparation referred to as DSS-W (71.4 g, 13.84%, w/w). DSS-A and DSS-W were mixed and concentrated to 1 g/ml, known as DSS extract, and stored at 4 °C.

JD-30 prepared from DSS-A. DSS-A was extracted with ethyl acetate followed by n-butanol; the products were designated DSS-A-E and DSS-A-N, respectively. The residue fraction was referred to as DSS-A-W. DSS-A-N (12.1 g, 2.34%, w/w) and DSS-A-W (28.2 g, 5.46%, w/w) were mixed and concentrated to remove solvent before being lyophilized to produce 3.575 g sample. The sample was dissolved in a sufficient quantity of distilled water and eluted in turn by distilled water, 10%, 30% and 95% ethanol on macroporous adsorptive resins (Diaion HP-20, Mitsubishi, Japan). A vacuum distillation technique was used to isolate and purify JD-30 from the 30% ethanol eluent, which produced a brown powder after cryodesiccation, and contained >80% saponins (26.69% albiflorin, 52.27% paeoniflorin and a small quantity of 4'-hydroxy-albiflorin, benzoylpaeoniflorin, lactiflorin, oxypaeoniflorin; Fig. 1). The yield of JD-30 was ~0.44% (w/w) in terms of DSS.

2.2. Animals

Original SAMR1 and SAMP8 mice were kindly provided by Dr. T. Takeda at Kyoto University, Japan. The mice were maintained in the Beijing Institute of Pharmacology and Toxicology under a 12 h light/12 h dark cycle at a constant temperature of 25 ± 1 °C, with a humidity of 55 ± 5%, and were fed a standard rodent diet. They were allowed free access to water and food. Management was carried out according to the care and use guide of laboratory animals by the NIH.

2.3. Groups and drug administration

Seven months SAMP8 mice were separated into 5 groups at random, each group of 20 mice (10 males and 10 females) were divided

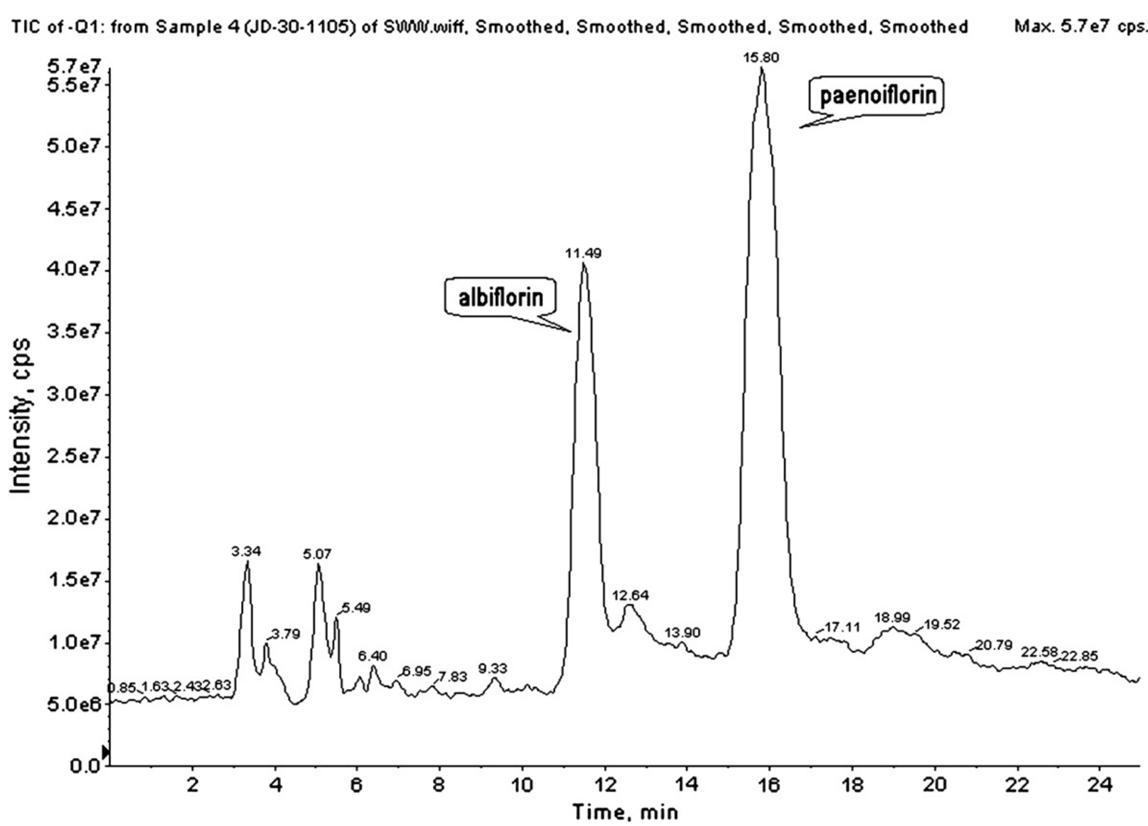

Fig. 1. Total ions chromatogram of JD-30 by high pressure liquid chromatography (Hu et al., 2010).

into 4 cages by gender. DSS was dissolved in distilled water at 320 mg/ml and JD-30 to 0.7, 1.4 and 2.8 mg/ml. The mice in drug-treated groups were given intragastric administration of DSS or JD-30 (0.1 ml/10 g body weight) once a day during the test. SAMP8 as negative control and the age-matched SAMR1 (10 males and 10 females) as positive control were given an equal volume of distilled water. The mice were weighed every 3 days. After 8 weeks of drug administration, the Morris water-maze test was performed and the drugs given lasted until the whole test was finished.

2.4. Morris water-maze task

The Morris water-maze test was performed 8 weeks after drug administration with minor modifications (D'Hooge and De Deyn, 2001). The tests were carried out between 8:00 and 12:00 AM. The water-maze was a circular pool 90 cm in diameter and 45 cm in height with a black inner surface. The tank was placed in a dim light sound-proof room with several visual cues. The pool was divided into 4 quadrants of equal area, and filled to a depth of 30 cm with water at 20 ± 2 °C. A black platform 6 cm in diameter and 1 cm below the surface of the water was placed in one of the quadrants.

Basic training included a hidden-platform acquisition training session and a probe trial session. The first test day was devoted to swimming training for 60 s without the submerged platform. During the 6 subsequent days, mice were given 3 daily trials in the presence of the platform. For 3 trial sessions, mice were placed into the water facing the pool wall in one of the quadrants. The group sequence was changed to a different order each day for data balance. When a mouse located the platform, it was permitted to remain on it for 10 s. If the mouse did not locate the platform within 60 s, it was placed on the platform for 10 s. The animals were taken to their cage and allowed to dry under a warm towel after each trial. During the each session, the time taken to find the hidden platform (escape latency) was recorded.

One day after the last training trial sessions, mice were subjected to a probe trial session in which the platform was removed from the pool. The mice were allowed to swim for 60 s to search for it. Latency (the first time that the mice crossed the former platform), searching times (the number of times that the mice crossed the former platform), and the swimming time within the target quadrant were recorded and analyzed.

2.5. Electrophysiology

Acute hippocampal slices were prepared from SAMR1 and SAMP8 (9 month old, female, 18–20 g). Under deep anesthesia with diethyl ether, the mouse was decapitated, and the hippocampal formation was rapidly resected and placed in ice-cold oxygenated (95% O_2 + 5% CO_2) artificial cerebrospinal fluid (ACSF, pH 7.2–7.4) containing in mM: 5 KCl, 124 NaCl, 2.5 $CaCl_2$, 1.3 $MgSO_4$, 1.2 KH_2PO_4, 26 $NaHCO_3$ and 10 D-$C_6H_{12}O_6$. Transverse slices (400 μm) were cut with a vibratome (Campden Instruments, Longhborough, U.K.) in ice-cold oxygenated ACSF.

Slices were incubated in ACSF (with or without drug) at 30 ± 2 °C for >2 h prior to electrophysiological recording. JD-30 was given at 25, 50 and 100 mg/L. Each concentration of drug was applied to an individual hippocampal slice to avoid cumulative effects. During experiments, an individual slice was transferred to a submersion-recording chamber and was continuously perfused with ACSF (1.5 ml/min) using a peristaltic pump (Minipuls2, Gilson, France) at 25 ± 1 °C.

Field population spike (PS) were recorded using Pclab 3.5 (MMStar Co., Beijing, China) by placing a glass pipette (3–5 MΩ) filled with NaCl (4 M) in the pyramidal layer of the hippocampal CA1 region. Stimuli (200 μs duration) were delivered at 0.017 HZ through a bipolar platinum electrode placed at the Schaffer collaterals from the CA3 region. Slices displaying an unstable baseline recording were discarded. After obtaining a stable baseline, stimulus–response curves for 30 min and the maximal PS amplitude were recorded with increasing stimulus intensity. The response curves evoked by the test stimulus pulse eliciting 50–60% of the maximum PS amplitude were recorded for 15 min before LTP was induced with the same stimulating strength by one train high frequency stimulus (HFS: 100 Hz, 100 pulses, 200 μs duration), and the PS was recorded 60 min later.

2.6. Brain collection of behaviorally-tested mice

The second day following completion of Morris water-maze test, all the mice were anesthetized with Nembutal (1 mg/10 g body weight). Six mice selected randomly were perfused through the ascending aorta with saline solution, followed by 4% paraformaldehyde in phosphate-buffered saline. The brains were removed immediately, fixed with 4% paraformaldehyde and embedded in paraffin. They were cut coronally into 4 μm slices using a vibratome (Leica RM2016), and the paraffin sections were prepared for Nissl staining, Aβ immunohistochemistry and Congo red staining to measure the neuron morphology, the content and the deposition of Aβ, respectively. Brains of other mice were removed, and the cortex and hippocampus were dissected. The 2 areas were immediately snap-frozen in liquid nitrogen and preserved for measurement of target gene expression and other possible index.

2.7. Nissl staining

Neuronal morphology in brain sections was evaluated after cresyl violet staining of Nissl bodies (Li et al., 2009). Briefly, after deparaffinage, sections were incubated for 10 min with cresyl violet (Sigma-Aldrich) solution, destained in 96% ethanol containing 0.5% acetic acid, dehydrated with isopropanol, cleared in xylene and mounted under coverslips.

A computer-assisted light microscope (OLYMPUS, BX51) was used to scan the sections. The densities of Nissl-positive cells in the hippocampal pyramidal cell layer (CA1, CA3 and CA4) and the dentate gyrus (DG) of the scanned digital images were calculated using Image-Pro Express software (Ver 5.1.1.14). The total cell counts were averaged from 5 sections on the 2 sides of the hippocampus of each animal.

2.8. Aβ immunohistochemistry

Immunohistochemical staining was preformed as described by Oddo et al. (2005). Briefly, deparaffinaged slides were boiled at 92–98 °C in 0.01 M citrate buffer (pH 6.0) for 15 min in a microwave oven to expose the epitope. Endogenous peroxidase activity was quenched for 10 min in 3% H_2O_2, and the sections were soaked for 20 min in 1% bovine serum albumin to block non-specific antigen attachment. The appropriate primary antibody (6E10, 1:1000, mouse monoclonal antibody against amino acids 1–17 of Aβ; Covane, SIG-39320) was applied, and sections were incubated overnight at 4 °C. Following 3 washes for 2 min in phosphate buffered solution to remove excess primary antibody, the sections were incubated in the appropriate secondary antibody (horseradish peroxidase-labeled coat anti-mouse IgG, 1:500; ZSJQ) for 1 h at room temperature. After a final washing in phosphate buffered saline, the antigen-antibody staining was visualized using diaminobenzidine. Finally, the sections were washed in distill water, dehydrated through an ethanol series, cleared with xylene and mounted in gel.

All sections were observed by light microscopy (OLYMPUS, BX51) and photo images were taken with Image-Pro Express software (Ver 5.1.1.14). Immunoreactive area and integrated optical density (IOD) served as the 2 main indices to quantify the effects of JD-30, using Image-Pro Plus software (Ver 6.0.0.260).

Table 1
The primer sequences of six genes for quantitative real-time PCR.

Gene	Forward primer	Reverse primer	Amplicon (bp)
GAPDH	CCTTCCGTGTTCCTACCC	CAACCTGGTCCTCAGTGTAG	150
APP	TTCGGACATGATTCAGGATTT	TGATGACAATCACGGTTGCTA	133
BACE1	AGGGCTTGCACCTGTAGGAC	GCCTGAGTATGACGCCAGTA	198
CatB	ACAAGCCTTCCTTCCACCCG	TGTCCTCACCGAACGCAACC	177
NEP	TCTTGTAAGCAGCCTCAGCC	CTCCCCACAGCATTCTCCAT	141
IDE	TGGTCATCTAATTGGGCACG	AACCTCGGGCTCCTTCCTTC	107

GAPDH, glyceraldehyde phosphate dehydrogenase; APP, amyloid protein precursor; BACE1, beta-site amyloid cleave enzyme-1; CatB, cathepsin B; NEP, neprilysin; IDE, insulin degrading enzyme.

2.9. Congo red staining

Detection of Aβ aggregates in the brain used Congo red staining (Rofina et al., 2004). Briefly, after deparaffinage sections were incubated for 10 min with hematoxylin solution, rinsed in tap water, and then incubated in Congo red (Sigma-Aldrich) solution for 20 min. The sections were washed in distill water, dehydrated through an ethanol series, cleared with xylene and mounted in gel. The sections were observed by light microscopy (OLYMPUS, BX51) and photos processed using Image-Pro Express software (Ver 5.1.1.14).

2.10. Preparation of total RNA and cDNA

Total RNA was extracted from hippocampus and cortex using Trizol reagent (Invitrogen) according to the manufacturer's instructions, and stored at −70 °C. The ratios of A_{260}/A_{280} in the range of 1.9–2.0 were used. RNA concentrations were determined from absorbance at 260 nm. Denaturing formaldehyde 1% agarose gel electrophoresis was used to check the integrity of total RNA. cDNA was synthesized from total RNA by reverse transcriptase (Promega) with the primer oligo $(dT)_{12–18}$ and was stored at −20 °C.

2.11. Relative quantitative real-time PCR

By using the primer sequences listed in Table 1, relative quantitative real-time PCR analysis was performed with an Applied Biosystems Prism 7500 Fast Sequence Detection System. The mRNA level of amyloid protein precursor (APP), beta-site amyloid cleave enzyme-1 (BACE1), cathepsin B (CatB), neprilysin (NEP) and insulin degrading enzyme (IDE) in cerebral cortex and hippocampus of all 6 groups were analyzed. Mouse GAPDH gene was used as endogenous control.

SYBR® Green Real-time PCR Master Mix (TOYOTO, QPK-201; for GAPDH, APP, BACE1 and CatB) and iTaq™ SYBR Green Supermix with Rox (BIO-RAD, 172–5851; for GAPDH, NEP and IDE) were used according to the manufacturer's instructions. The thermal cycler conditions were as follows: hold for 120 s at 95 °C, followed by three-step PCR for 40 cycles of 95 °C for 15 s, 58 °C (GAPDH, APP, BACE1 and CatB) or 60 °C (GAPDH, NEP and IDE) for 30 s, and 72 °C for 45 s.

Levels of mRNA expression were determined using the 7500 Fast System SDS software version 1.3.1 (Applied Biosystems) according to the $2^{-\triangle\triangle Ct}$ method. Briefly, quantitative data representing the mRNA expression level of genes derived from the standard curve of threshold cycle (Ct) values were normalized to the internal control GAPDH relative to a calibrator, consisting of the mean expression level of the corresponding gene in control samples as follows: $2^{-\triangle\triangle Ct} = 2^{-((\text{Ct test gene} - \text{Ct GAPDH})\ \text{sample} - (\text{Ct test gene} - \text{Ct GAPDH})\ \text{calibrator})}$. The results came from different cDNA from 4 mice.

2.12. Statistical analysis

All data are expressed as mean ± SEM. Origin 7.5 (Originlab Co., USA) and SAS software package (Release 8.01, Sas Institute Inc, USA) were used to plot and analyze data by Student's *t*-test for 2 groups, and one-way analysis of variance (ANOVA) was used for >2 groups, followed by a Dunnett post-hoc test. $P<0.05$ was taken as statistically significant.

3. Results

3.1. JD-30 improved spatial learning and memory deficits

In the Morris water-maze training session, SAMR1 rapidly learned the location of the platform, but SAMP8 had significant longer escape latency on every test days. This increase was significantly shortened by JD-30 (7, 14, 28 mg/kg) administration from 2 to 6 day, and by DSS (3.2 g/kg) administration on only the 3rd day (data in Table 2).

Similarly, on the day of the probe trial following the final day of training, latency (the first time that the mice crossed the former platform) was increased (Fig. 2A) and searching times (the number of times that the mice crossed the former platform) was decreased more in SAMP8 than in SAMR1(Fig. 2B). The latency was decreased by JD-30 or DSS administration (Fig. 2A), and the searching times were increased significantly by high dosage JD-30 administration (Fig. 2B). Although JD-30 or DSS could not increase the shortened swimming time within the target quadrant (Fig. 2C), the above results indicate that JD-30 improves spatial learning and memory deficits in SAMP8.

3.2. JD-30 elevated LTP reduction in hippocampal slices

A significant decrease in the average PS amplitude after HFS of SAMP8 (103.0% ± 3.8%) was observed compared with SAMR1 (132.0% ± 3.6%). For the slices of SAMP8 exposed to JD-30, the average PS amplitude after HFS was 113.5% ± 4.2% for 25 mg/L, 128.4% ± 7.3% for 50 mg/L, and 143. 6% ± 21.49% for 100 mg/L (Fig. 3).

Table 2
Effects of JD-30 on spatial learning and memory ability in SAMP8 during the training trial sessions of the Morris water maze performance.

Day	Escape latency (s)					
	SAMR1	SAMP8	DSS (g/kg)	JD-30 (mg/kg)		
			3.2	7	14	28
1	30.86 ± 2.25	42.45 ± 2.46**	38.25 ± 3.14	37.30 ± 2.29	38.96 ± 2.50	36.26 ± 3.44
2	21.44 ± 1.87	46.69 ± 2.75**	41.51 ± 3.10	37.59 ± 3.97	34.07 ± 2.99#	33.47 ± 3.93##
3	18.70 ± 1.92	48.00 ± 2.84**	38.64 ± 3.52#	28.56 ± 2.76##	39.05 ± 3.31#	36.23 ± 3.93##
4	21.48 ± 2.00	46.07 ± 3.09**	38.58 ± 3.16	37.66 ± 3.29	36.30 ± 2.73#	36.21 ± 3.28#
5	19.02 ± 2.33	47.31 ± 3.44**	39.19 ± 3.22	36.87 ± 3.39#	32.33 ± 3.35#	34.20 ± 4.01#
6	22.78 ± 1.79	47.91 ± 2.58**	40.18 ± 3.60	36.40 ± 3.67#	33.84 ± 3.70#	36.39 ± 2.45##

This table lists the data of the latency time during the training trial sessions of the Morris water maze performance. Spatial learning and memory ability of SAMP8 was impaired all throughout the test; and the impairment was improved by the administration of different dosages JD-30 or DSS. Data values are expressed as mean ± SEM, n = 20. ** $P<0.01$, compared with SAMR1; # $P<0.05$, ## $P<0.01$, compared with SAMP8.

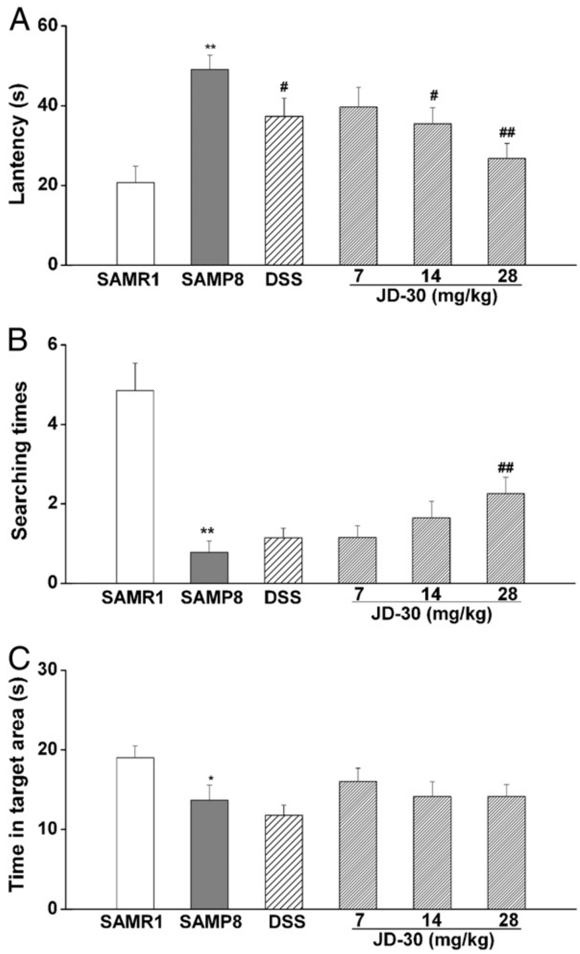

Fig. 2. Effects of JD-30 on spatial learning and memory ability during the probe trial session of the Morris water-maze performance in SAMP8. (A) Latency, the first time that the mice crossed the former platform. (B) Searching times, the number of times that the mice crossed the former platform. (C) The swimming time within the target quadrant. Data values are expressed as mean ± SEM, n = 20. $^{*}P<0.05$, $^{**}P<0.01$, compared with SAMR1; $^{\#}P<0.05$, $^{\#\#}P<0.01$, compared with SAMP8.

Compared with SAMP8, 50 and 100 mg/L JD-30 significantly elevated the PS amplitude. The results indicate that JD-30 elevates LTP reduction in SAMP8 hippocampal slices.

3.3. JD-30 increased the Nissl-positive cell densities in hippocampus

Intact cells were smoothly stained by Nissl into a light blue color with regularly shaped cell bodies and large centrally located nuclei. This stain allowed us to see the pyramidal neuron layer of CA1, CA3 and CA4 region and the granule neuron of dentate gyrus (DG) in hippocampus. Quantification of Nissl-positive cell densities (positive cells per 0.01 mm^2) in the hippocampus was shown in Fig. 4. Cell densities in the CA1 and CA3 regions showed no significant differences among the groups. While the cell densities in the CA4 region and DG were significantly decreased in SAMP8 compared with SAMR1, JD-30 significantly prevented cell loss in the hippocampus after long-term administration.

In the CA4 region and DG of SAMP8, there was loss accompanying neuronal swollen and abnormal arrangement (Fig. 5). In contrast, neurons in other groups were intact and well-arranged. The results indicate that long-term administration of JD-30 protects hippocampal neurons from damage in the brain of SAMP8.

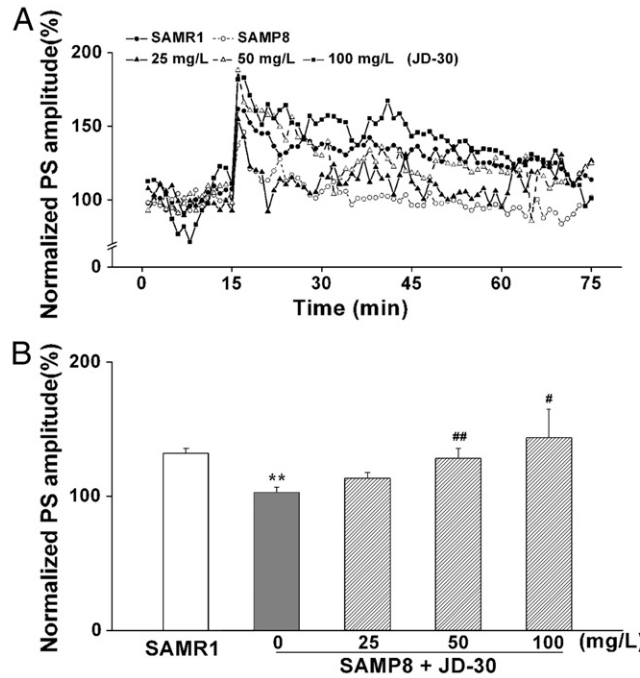

Fig. 3. Effects of JD-30 on LTP reduction in SAMP8 hippocampal slice. (A) PS amplitude and LTP induction were recorded in the CA1 region of mice hippocampal slices in different groups. (B) Summary of mean PS amplitude (16–75 min) from all experiments shown in A. Data values are expressed as mean ± SEM, n = 3-8. $^{**}P<0.01$, compared with SAMR1; $^{\#}P<0.05$, $^{\#\#}P<0.01$, compared with SAMP8. Single HFS was given at 16 min and recorded for 60 min. The mean PS amplitude before HFS (1 to 15 min) was normalized to 100%, and the PS amplitude was normalized to it at each time-point.

3.4. JD-30 decreased the content and deposition of Aβ in the brain

Immunohistochemical staining showed that the content of Aβ significantly increased in the brain of SAMP8, mainly in the cortex (Fig. 6C–D) compared with age-matched SAMR1 (Fig. 6A–B). Long-term administration of JD-30 or DSS significantly decreased the Aβ content of SAMP8 (Fig. 6E–L). Quantitatively a significant increase in Aβ immunoreactive area and integrated optical density (IOD) in SAMP8 was seen (Fig. 7); JD-30 significantly decreased the area and IOD in much the same way as DSS.

Aβ deposition in the brain of SAM was examined by Congo red staining. Deposition of Aβ of SAMP8 significantly increased (the parietal cortex, cortex of temporal lobe; CA3 region, CA4 region and dentate gyrus of hippocampus) compared with age-matched SAMR1 (Fig. 8). JD-30 or DSS administration significantly decreased Aβ

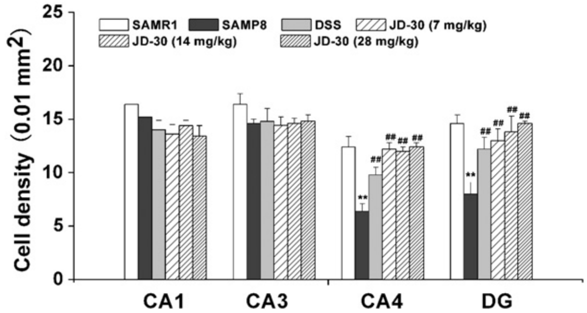

Fig. 4. Effects of JD-30 on the Nissl-positive cell densities (positive cells per 0.01 mm^2) in the CA1, CA3 and CA4 regions and DG of hippocampus. Data values are expressed as mean ± SEM, n = 5. $^{**}P<0.01$, compared with SAMR1; $^{\#\#}P<0.01$, compared with SAMP8.

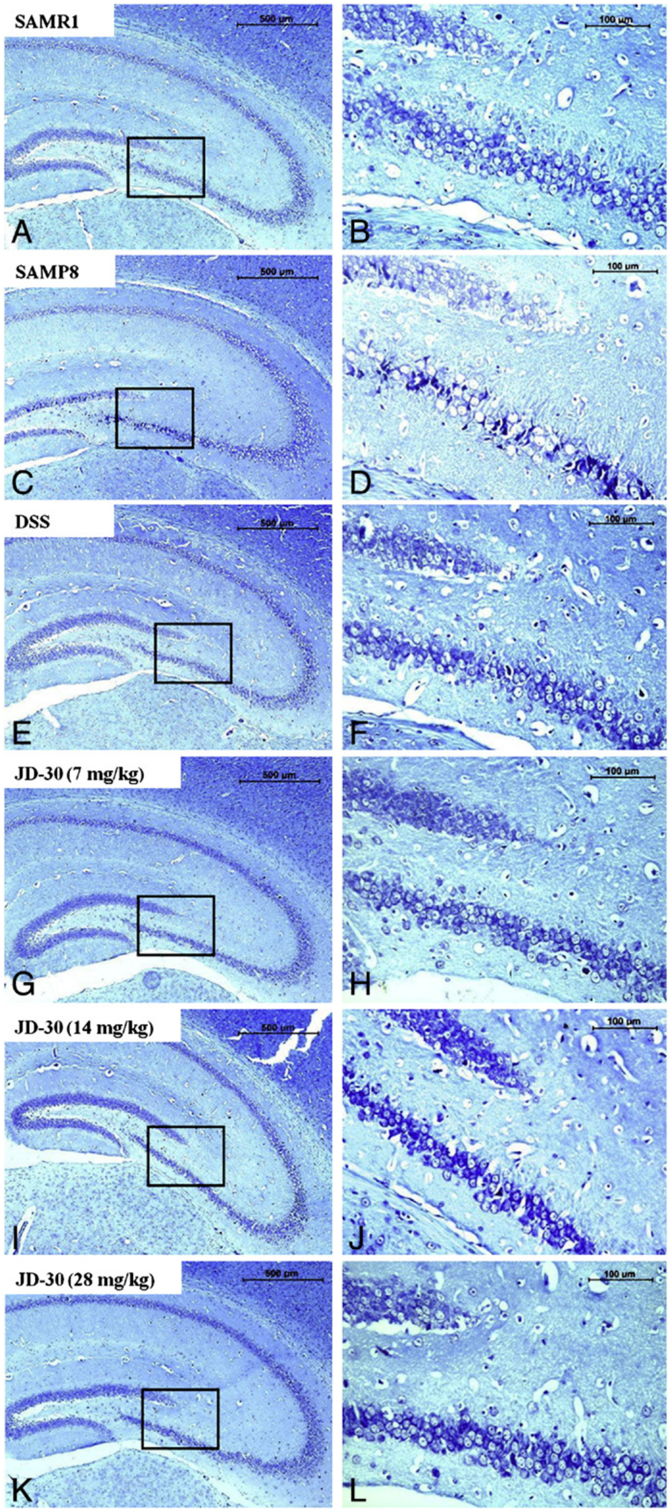

Fig. 5. Effects of JD-30 on the Nissl-positive cell densities in the hippocampus of the mice. SAMR1 showed intact and well-arranged neurons (A–B); in contrast, neurons of SAMP8 showed reduced staining, and they were significantly swollen and arranged abnormally (C–D). Long-term administration of JD-30 (G–L) or DSS (E–F) diminished the loss and swelling of neurons in the hippocampus, and neurons recovered their characteristic shape and arrangement. Scale bar: 500 μm for A, C, E, G, I, K and 100 μm for B, D, F, H, J, L.

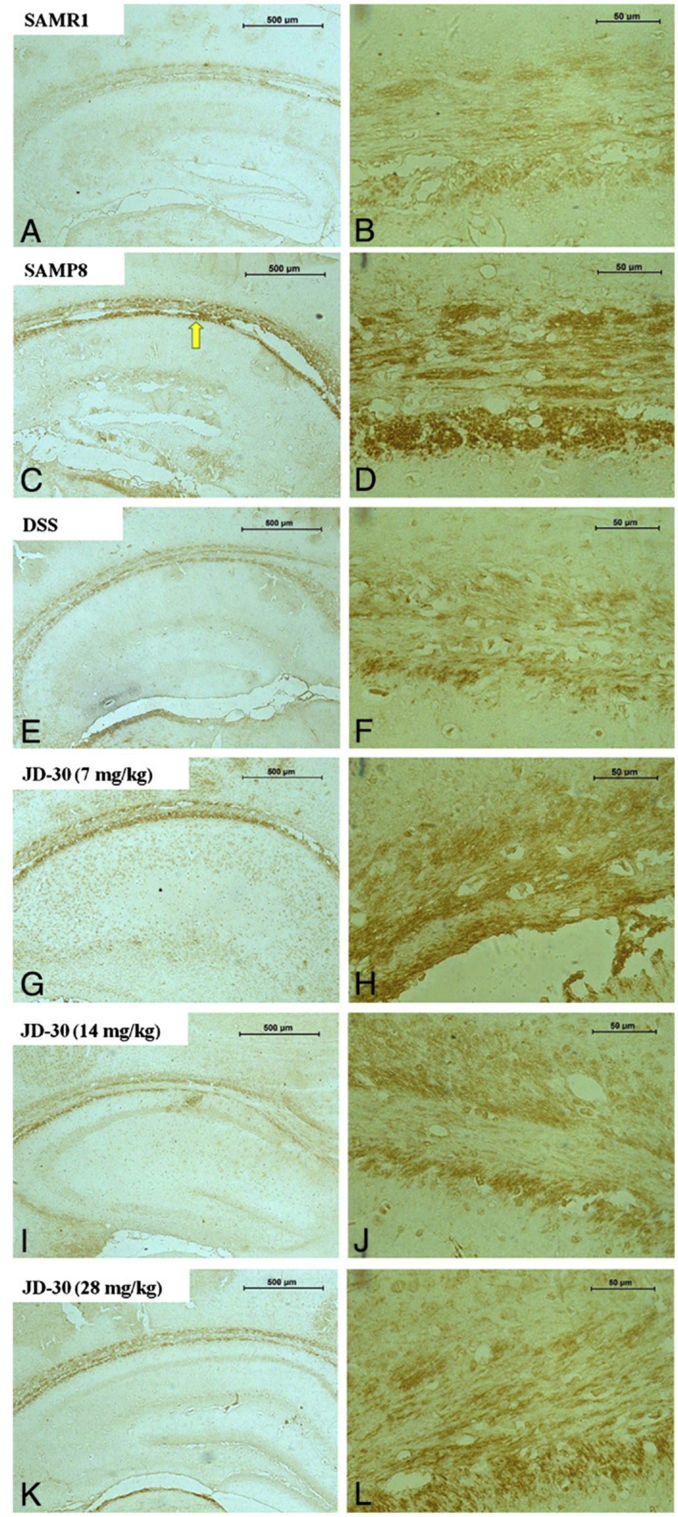

Fig. 6. Effects of JD-30 on the Aβ content in the brain of SAMP8 determined by immunohistochemical staining. The content of Aβ in SAMP8 (C–D) brain significantly increased (mainly in the cortex, yellow arrow) compared with age-matched SAMR1 (A–B). Long-term administration of JD-30 (G–L) or DSS (E–F) significantly decreased the content of Aβ in SAMP8. Scale bar: 500 μm for A, C, E, G, I, K and 50 μm for B, D, F, H, J, L.

deposition, especially in the CA3 and CA4 regions of the hippocampus. The results indicate that long term administration of JD-30 decreases the content and deposition of Aβ in the brain of SAMP8.

3.5. Effects of JD-30 on the expression of genes related with Aβ

mRNA levels of APP, BACE1, CatB, NEP and IDE in the cerebral cortex and hippocampus of all 6 groups were analyzed real-time PCR. Expression of these genes was not significantly different between SAMR1 and SAMP8 (Fig. 9).

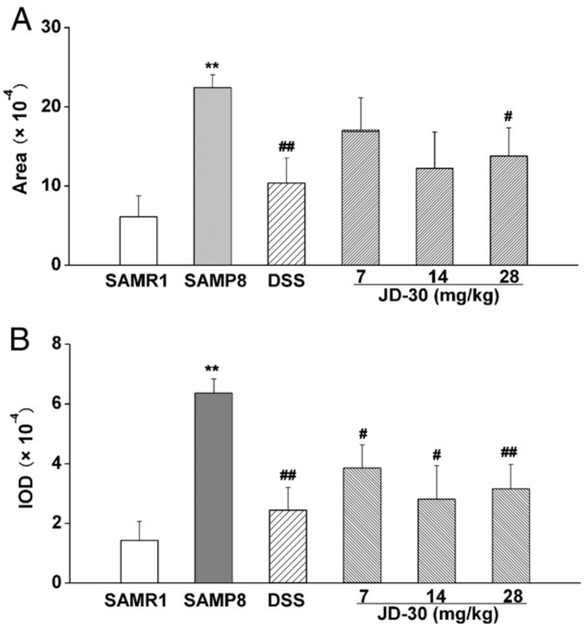

Fig. 7. Quantitative analysis of the effects of JD-30 on the Aβ content in the brain of SAMP8. (A) Aβ immunoreactive area. (B) IOD, integrated optical density. Data values are expressed as mean ± SEM, n = 5–7. $^{**}P<0.01$, compared with SAMR1; $^{\#}P<0.05$, $^{\#\#}P<0.01$, compared with SAMP8.

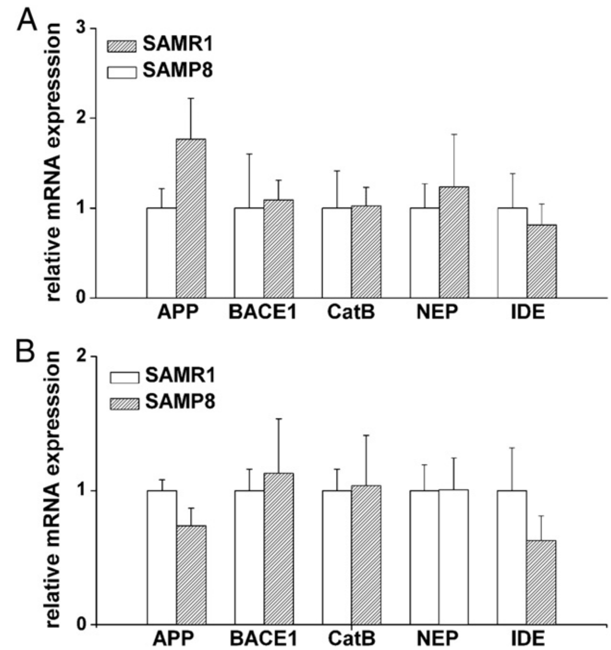

Fig. 9. APP, BACE1, CatB, NEP and IDE mRNA expression were assayed by real-time PCR in the cerebral cortex (A) and the hippocampus (B) of SAMP8 and SAMR1. Compared with age-matched SAMR1, APP mRNA was increased in the cortex, and IDE mRNA was decreased in the hippocampus of SAMP8, but the differences are not statistically significant. Data values are expressed as mean ± SEM, n = 4.

Cortical APP mRNA was increased in SAMP8 and hippocampal IDE mRNA was to an extent decreased in SAMP8. Looking at the effects of JD-30 on the expression of these 2 genes, JD-30 or DSS administration

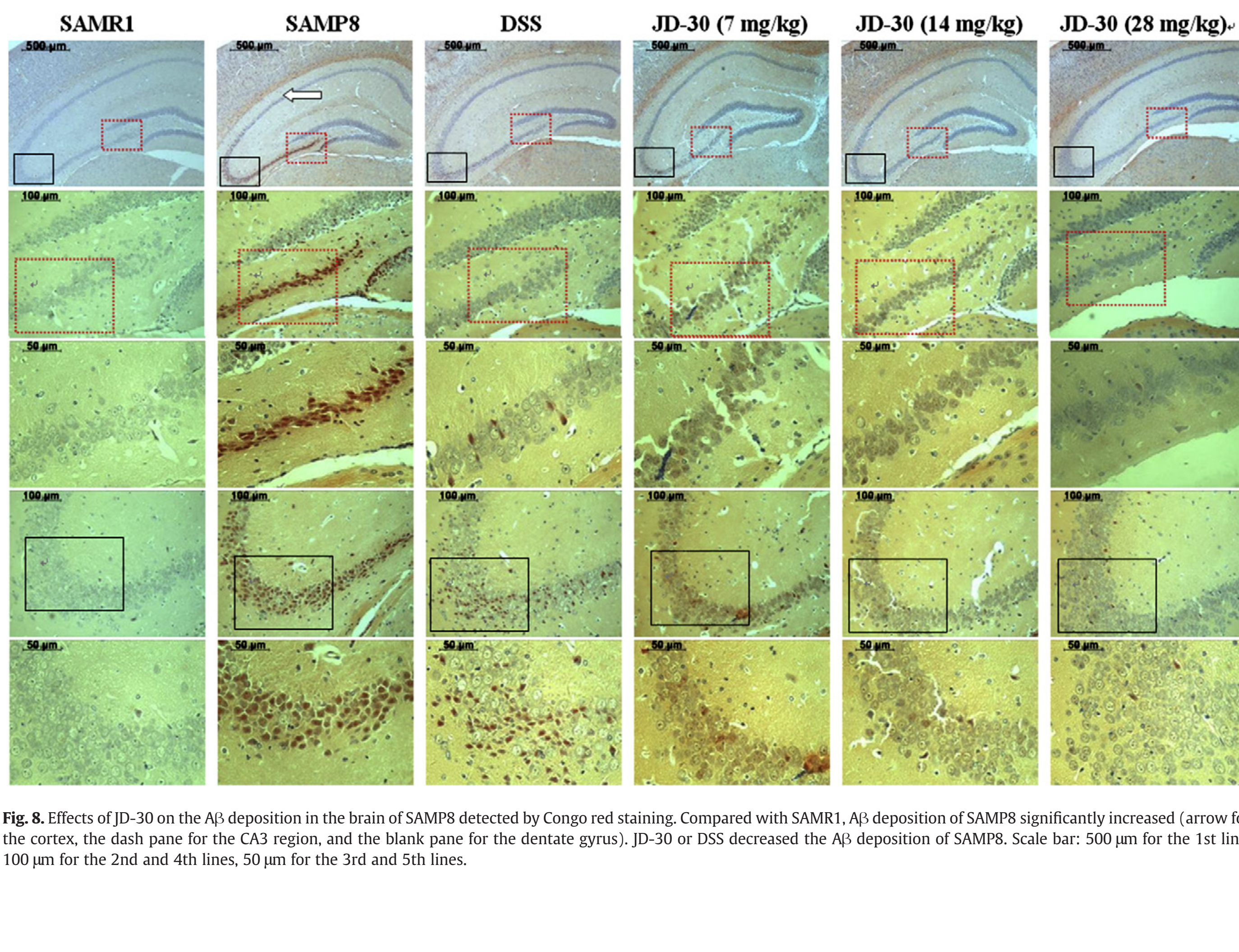

Fig. 8. Effects of JD-30 on the Aβ deposition in the brain of SAMP8 detected by Congo red staining. Compared with SAMR1, Aβ deposition of SAMP8 significantly increased (arrow for the cortex, the dash pane for the CA3 region, and the blank pane for the dentate gyrus). JD-30 or DSS decreased the Aβ deposition of SAMP8. Scale bar: 500 μm for the 1st line, 100 μm for the 2nd and 4th lines, 50 μm for the 3rd and 5th lines.

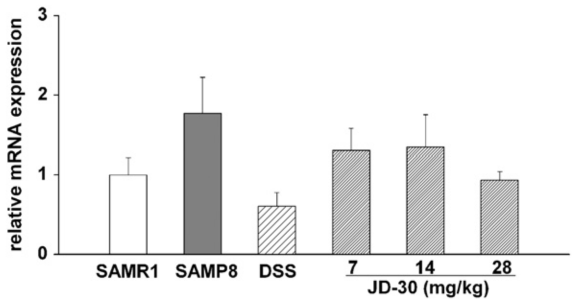

Fig. 10. Effects of JD-30 on the expression of cortical APP mRNA of SAMP8. Compared with age-matched SAMR1, cortical APP mRNA of SAMP8 was increased; JD-30 or DSS administration lowered cortical APP mRNA expression, but the differences are not statistically significant. Data values are expressed as mean ± SEM, n = 4.

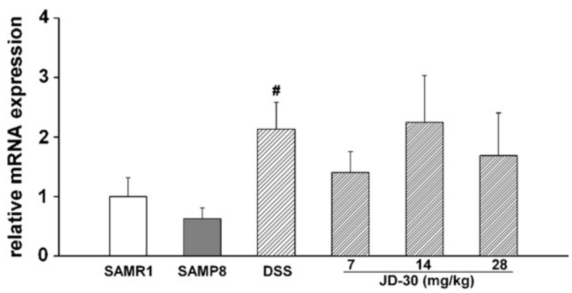

Fig. 11. Effects of JD-30 on the expression of hippocampal IDE mRNA of SAMP8. Compared with age-matched SAMR1, hippocampal IDE mRNA of SAMP8 was decreased; JD-30 or DSS administration groups had higher hippocampal IDE mRNA expression, but most of differences were not statistical significant. Data values are expressed as mean ± SEM, n = 4. $^{\#}P<0.05$, compared with SAMP8.

down-regulated cortical APP mRNA (Fig. 10) and up-regulate hippocampal IDE mRNA (Fig. 11), but the effects were small and not significantly statistical differences.

4. Discussion

There is at present no cure for AD. Although several drugs (donepezil, galantamine, rivastigmine and memantine) have been approved by the Food and Drug Administration of the US, the therapeutic effect is limit. Lots of researches aimed at improving medication, but with little success (Mangialasche et al., 2010; Neugroschl and Sano, 2010). Owing to the limitations and drawback of orthodox medicine and nutritional supplements, some patients resort to herbal medications (Man et al., 2008). Traditionally, a number of herbs have been used for cognitive disorders. For example, *Codonopsis pilulosa root* (Dang shen), *Biota orientalis* (Ce bai ye) and *Polygala tenuifolia root* (Yuan zhi) are used for amnesia in traditional Chinese medicine. Some compounds with cognition-improving properties have been isolated from plants, such as EGb761 (extract from *Ginkgo biloba*), Huperzine A (extract from *Lycopodium serratum*).

DSS can ameliorate cognitive deterioration caused by aging, cerebral ischemia and chemical injury to the brain (Hatip-Al-Khatib et al., 2004; Mizushima et al., 2003; Pu et al., 2005), but the underlying mechanisms and substance basis of DSS on AD remain elusive. Extracting bioactive fractions from the prescribed Chinese herbs is a valid approach from which substrates and effects can be assessed pharmacologically. After extracting several fractions from DSS, preliminary studies had indicated that JD-30 may be an active fraction because it had good neuroprotective action, better quality control and a relative clear ingredient.

SAM was originally developed from the AKR/J strain mice in 1968 in the laboratory of Professor Toshio Takeda in Kyoto, Japan, based on the data of the grading score of senescence, life span and pathologic phenotypes (Nomura et al., 1996; Nomura and Okuma, 1999). The SAMP8 subline is characterized by early onset of deficits in learning and memory, cholinergic deficit in the hippocampus,age-related increase in Aβ-like deposition and amyloid plaques (Chen et al., 2004; Miyamoto, 1997; Takeda, 1999). A comparison of the properties of SAMP8 and the characteristic features of AD shows some similarities, suggesting that SAMP8 serves as a good animal model to investigate the fundamental mechanisms of AD and assess the action of drugs (Okuma and Nomura, 1998; Takeda, 2009). There has been no study regarding JD-30 on these deficiencies in SAMP8, which became the purpose of our research

Long-term administration of JD-30 ameliorated the behavior impairment of SAMP8, and JD-30 was superior in the elevation of spatial learning and memory than DSS. Consistent with the behavioral data, the electrophysiology experiments showed that JD-30 repaired LTP reduction in hippocampal slices of SAMP8. Additionally, our previous studies had shown that JD-30 could also improve the impairment of spatial cognition and LTP induction induced by $A\beta_{25-35}$ (Hu et al., 2007, 2010). These results suggest that JD-30 is one of the chief active fractions of DSS for improving cognition deterioration, and that reversing the reduction of LTP may provide a neurobiology basis for improving the deterioration of spatial recognition abilities in the mice.

The abnormal structure and neuronal loss in the hippocampus of SAMP8 might be linked to cognitive dysfunction and synaptic plasticity reduction. Administration of JD-30 protected hippocampal neurons from damage, and the cells recovered their characteristic shape and arrangement. The result indicates that protecting the intact of neuron and synapse in hippocampus may be the anatomical basis for restoring the spatial cognition and synaptic plasticity deficits.

Aggregation and deposition of Aβ in the brain is a key step in the pathogenesis of AD, which elicits a cascade of cellular events leading ultimately to neuronal loss and dementia (Hardy and Selkoe, 2002). We found that the content and deposition of Aβ in the brain of SAMP8 significantly increased compared with age-matched SAMR1; the increase may be contributed to the acceleration of senescence, learning and memory deficit. Long-term administration of JD-30 significantly decreased the content and deposition of Aβ. Meanwhile, our previous findings had shown that JD-30 not only blocked the aggregation of Aβ, but disrupted aggregated Aβ *in vitro* (Hu et al., 2010). JD-30 could decrease the content, aggregation and deposition of Aβ, which may be one of the mechanisms of action of JD-30 on synaptic plasticity dysfunction and cognition degeneration in the AD model mice. Reducing the production of Aβ and increasing its clearance, followed by reducing Aβ-mediated neuronal damage, might be an optimal intervention in AD treatment (Aisen, 2005). Our findings indicate that extracting the active fractions or compounds from Chinese herbs is a feasible method for AD treatment.

Imbalance between the production of Aβ and its removal causes Aβ accumulation. Aβ is produced by proteolytic processing of APP. Proteases referred to as β-secretase and γ-secretase cleave at the N- and C-termini of Aβ within APP to generate Aβ. Aβ then undergoes secretion to provide extracellular Aβ that produces neurotoxic effects, with aggregation and accumulation in amyloid plaques of AD brains. BACE1 and CatB are generally considered to be the critical β-secretases (Hook et al., 2008; Haque et al., 2008; Stockley and O'Neill, 2008). Additionally, several enzymes within the central nervous system are capable of degrading Aβ (Malito et al., 2008; Miners et al., 2008). Among these enzymes, there is increasing evidence that NEP and IDE are mostly responsible for degrading Aβ in the brain (Chen et al., 2010). On this basis, 5 genes associated with Aβ pathway, including APP, BACE1, CatB, NEP and IDE, were analyzed to explore the molecular basis of the abnormal Aβ level, but none were significantly different in SAMR1 and SAMP8. Although JD-30 down-regulated cortical APP mRNA and up-regulated hippocampal IDE mRNA to a certain extent, the effects were limited. But this study examined only the mRNA expression of relative molecule, and did not investigate protein expression and enzyme activity. Notwithstanding its limitation, our present study suggests that JD-30 decreases the content and deposition of Aβ in the brain, which might not be due to regulation of gene expression, but influences protein translation and/or enzyme activity. Further studies are needed.

We have demonstrated that JD-30 improved spatial learning and memory deficits in SAMP8, reversed the LTP reduction in hippocampal slices, protected the hippocampal neuron from damage, and decreased the content and deposition of Aβ. JD-30 is therefore one of the active fractions of DSS. JD-30's neuroprotective action against age-related senescence makes it a potential therapeutic agent for AD treatment.

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References

Aisen, P.S., 2005. The development of anti-amyloid therapy for Alzheimer's disease from secretase modulators to polymerisation inhibitors. CNS Drugs 19, 989–996.
Alzheimer's Association, 2009. Alzheimer's disease facts and figures. Alzheimers Dement. 5, 234–270.
Butterfield, D.A., Poon, H.F., 2005. The senescence-accelerated prone mouse (SAMP8): a model of age-related cognitive decline with relevance to alterations of the gene expression and protein abnormalities in Alzheimer's disease. Exp. Gerontol. 40, 774–783.
Chen, G.H., Wang, Y.J., Wang, X.M., Zhou, J.N., 2004. Accelerated senescence prone mouse-8 shows early onset of deficits in spatial learning and memory in the radial six-arm water maze. Physiol. Behav. 82, 883–890.
Chen, P.T., Liao, T.Y., Hu, C.J., Wu, S.T., Wang, S.S., Chen, R.P., 2010. A highly sensitive peptide substrate for detecting two Abeta-degrading enzymes: neprilysin and insulin-degrading enzyme. J. Neurosci. Methods 190, 57–62.
Cummings, J.L., 2004. Alzheimer's disease. N. Engl. J. Med. 351, 56–67.
Del Valle, J., Duran-Vilaregut, J., Manich, G., Casadesús, G., Smith, M.A., Camins, A., et al., 2010. Early amyloid accumulation in the hippocampus of SAMP8 mice. J. Alzheimers. Dis. 19, 1303–1315.
D'Hooge, R., De Deyn, P.P., 2001. Applications of the Morris water maze in the study of learning and memory. Brain Res. Brain Res. Rev. 36, 60–90.
Fujii, T., 2002. Herbal factors in the treatment of autoimmunity-related habitual abortion. Vitam. Horm. 65, 333–344.
Hagino, N., 1994. An overview of Kampo Medicine: Toki-Shakuyaku-San (TJ-23). Phytother. Res. 7, 391–394.
Haque, A., Banik, N.L., Ray, S.K., 2008. New Insights into the Roles of Endolysosomal Cathepsins in the Pathogenesis of Alzheimer's disease: Cathepsin Inhibitors as Potential Therapeutics. CNS Neurol. Disord. Drug. Targets 7, 270–277.
Hardy, J., Selkoe, D.J., 2002. The amyloid hypothesis of Alzheimer's disease: progress and problems on the road to therapeutics. Science 297, 353–356.
Hatip-Al-Khatib, I., Egashira, N., Mishima, K., Iwasaki, K., Iwasaki, K., Kurauchi, K., et al., 2004. Determination of the effectiveness of components of the herbal medicine Toki-Shakuyaku-San and fractions of Angelica acutiloba in improving the scopolamine-induced impairment of rat's spatial cognition in eight-armed radial maze test. J. Pharmacol. Sci. 96, 33–41.
Higaki, S., Toyomoto, T., Morohashi, M., 2002. Seijo-bofu-to, Jumi-haidoku-to and Toki-shakuyaku-san suppress rashes and incidental symptoms in acne patients. Drugs Exp. Clin. Res. 28, 193–196.
Hook, V., Schechter, I., Demuth, H.U., Hook, G., 2008. Alternative pathways for production of beta-amyloid peptides of Alzheimer's disease. Biol. Chem. 389, 993–1006.
Hu, Z.Y., Yang, Y., Zhou, W.X., Zhang, Y.X., Shang, W.W., Qiao, S.Y., 2007. JD-30, an active constituent extracted from Danggui Shaoyao San, ameliorates amyloid beta-protein fragment 25–35 induced inhibition of long-term potentiation of CA1 area in rat hippocampal slices. Chin. J. Pharmacol. Toxicol. 21, 185–189.
Hu, Z.Y., Liu, G., Yuan, H., Yang, S., Zhou, W.X., Zhang, Y.X., et al., 2010. Danggui–Shaoyao–San and its active fraction JD-30 improve Aβ-induced spatial recognition deficits in mice. J. Ethnopharmacol. 128, 365–372.
Inanaga, K., 2007. Aging in brain and mind-drug therapy for patients with behavioral and psychological symptoms of dementia (BPSD). Seishin Shinkeigaku Zasshi 109, 703–708.
Kotani, N., Oyama, T., Sakai, I., Hashimoto, H., Muraoka, M., Ogawa, Y., et al., 1997. Analgesic effect of a herbal medicine for treatment of primary dysmenorrhea—a double-blind study. Am. J. Chin. Med. 25, 205–212.
Li, Q., Zhao, H.F., Zhang, Z.F., Liu, Z.G., Pei, X.R., Wang, J.B., et al., 2009. Long-term green tea catechin administration prevents spatial learning and memory impairment in senescence-accelerated mouse prone-8 mice by decreasing Abeta1-42 oligomers and upregulating synaptic plasticity-related proteins in the hippocampus. Neuroscience 163, 741–749.
Malito, E., Hulse, R.E., Tang, W.J., 2008. Amyloid beta-degrading cryptidases: insulin degrading enzyme, presequence peptidase, and neprilysin. Cell. Mol. Life Sci. 65, 2574–2585.
Man, S.C., Durairajan, S.S., Kum, W.F., Lu, J.H., Huang, J.D., Cheng, C.F., et al., 2008. Systematic review on the efficacy and safety of herbal medicines for Alzheimer's disease. J. Alzheimers. Dis. 14, 209–223.
Mangialasche, F., Solomon, A., Winblad, B., Mecocci, P., Kivipelto, M., 2010. Alzheimer's disease: clinical trials and drug development. Lancet Neurol. 9, 702–716.
Miners, J.S., Baig, S., Palmer, J., Palmer, L.E., Kehoe, P.G., Love, S., 2008. Abeta-degrading enzymes in Alzheimer's disease. Brain Pathol. 18, 240–252.
Miyamoto, M., 1997. Characteristics of age-related behavioral changes in senescence-accelerated mouse SAMP8 and SAMP10. Exp. Gerontol. 32, 139–148.
Mizushima, N., 1989. Effects of TJ-23 (Toki-shakuyaku-san) on senile dementia. Jpn. Med. Pharm. Soc. Wakan-Yaku 6, 456–457.
Mizushima, Y., Kan, S., Yoshida, S., Irie, Y., Urata, Y., 2003. Effect of Choto-san, a Kampo medicine, on impairment of passive avoidance performance in senescence accelerated mouse (SAM). Phytother. Res. 17, 542–545.
Neugroschl, J., Sano, M., 2010. Current Treatment and Recent Clinical Research in Alzheimer's Disease. Mt. Sinai J. Med. 77, 3–16.
Nomura, Y., Okuma, Y., 1999. Age-related defects in lifespan and learning ability in SAMP8 mice. Neurobiol. Aging 20, 111–115.
Nomura, Y., Yamanaka, Y., Kitamura, Y., Arima, T., Ohnuki, T., Oomura, Y., et al., 1996. Senescence-accelerated mouse. Neurochemical studies on aging. Ann. N. Y. Acad. Sci. 786, 410–418.
Oddo, S., Caccamo, A., Green, K.N., Liang, K., Tran, L., Chen, Y., et al., 2005. Chronic nicotine administration exacerbates tau pathology in a transgenic model of Alzheimer's disease. Proc. Natl. Acad. Sci. U. S. A. 102, 3046–3051.
Okuma, Y., Nomura, Y., 1998. Senescence-accelerated mouse (SAM) as an animal model of senile dementia: pharmacological, neurochemical and molecular biological approach. Jpn. J. Pharmacol. 78, 399–404.
Pu, F., Mishima, K., Egashira, N., Akiyoshi, Y., Liu, A.X., Sano, K., et al., 2005. Post-ischemic treatment with toki-shakuyaku-san (tang-gui-shao-yao-san) prevents the impairment of spatial memory induced by repeated cerebral ischemia in rats. Am. J. Chin. Med. 33, 475–489.
Rofina, J.E., Singh, K., Skoumalova-Vesela, A., van Ederen, A.M., van Asten, A.J., Wilhelm, J., et al., 2004. Histochemical accumulation of oxidative damage products is associated with Alzheimer-like pathology in the canine. Amyloid 11, 90–100.
Stockley, J.H., O'Neill, C., 2008. Understanding BACE1: essential protease for amyloid-beta production in Alzheimer's disease. Cell. Mol. Life Sci. 65, 3265–3289.
Takeda, T., 1999. Senescence-accelerated mouse (SAM): a biogerontological resource in aging research. Neurobiol. Aging 20, 105–110.
Takeda, T., 2009. Senescence-accelerated mouse (SAM) with special references to neurodegeneration models, SAMP8 and SAMP10 mice. Neurochem. Res. 34, 639–659.
Yagi, H., Katoh, S., Akiguchi, I., Takeda, T., 1988. Age-related deterioration of ability of acquisition in memory and learning in senescence accelerated mouse: SAM-P/8 as an animal model of disturbances in recent memory. Brain Res. 474, 86–93.